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CAPACITIVE JUNCTION BETWEEN CONDUCTIVE LINE AND CONDUCTIVE PILLAR WITH METHODS TO FORM SAME

Abstract

Structures of the disclosure include a first conductive line within a dielectric material. The dielectric material extends over the first conductive line. A second conductive line is within the dielectric material and substantially vertically aligned with the first conductive line. A conductive pillar is within the dielectric material between the first conductive line and the second conductive line. The conductive pillar includes an upper surface contacting a lower surface of the second conductive line or a lower surface contacting an upper surface of the first conductive line. A vertical thickness of the conductive pillar is less than a vertical thickness between the first conductive line and the second conductive line. a first capacitive junction is between the conductive pillar and one of the first conductive line and the second conductive line. A second capacitive junction is between the conductive pillar and a horizontally adjacent conductive pillar.

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Background/Summary

BACKGROUND

[0001] The present disclosure relates to integrated circuit structures and, more particularly, to conductive lines of integrated circuit structures.

[0002] Capacitors are used in a wide variety of integrated circuits (ICs). As ICs have scaled smaller, forming capacitors has become more challenging. Although it is possible to form capacitors in the metal wiring layers of a device (i.e., the “back end of line” (BEOL) layers), these structures conventionally will occupy significant chip area to achieve desired amounts of capacitance. Some implementations of capacitors in metal wiring layers may require, e.g., forming additional, dedicated capacitor dielectric layers for the various capacitor devices to operate. Such techniques undesirably increase the number of masks and/or processing phases to form a device.

SUMMARY

[0003] All aspects, examples and features mentioned below can be combined in any technically possible way.

[0004] An aspect of the disclosure provides a structure including: a first conductive line within a dielectric material, the dielectric material extending over the first conductive line; a second conductive line within the dielectric material and substantially vertically aligned with the first conductive line; and a conductive pillar within the dielectric material between the first conductive line and the second conductive line, wherein the conductive pillar includes one of an upper surface contacting a lower surface of the second conductive line and a lower surface contacting an upper surface of the first conductive line, wherein a vertical thickness of the conductive pillar is less than a vertical thickness between the first conductive line and the second conductive line, a first capacitive junction is between the conductive pillar and one of the first conductive line and the second conductive line, and a second capacitive junction is between the conductive pillar and a horizontally adjacent conductive pillar.

[0005] An aspect of the disclosure includes a structure, including: a first plurality of conductive lines within a dielectric material, each of the first plurality of conductive lines having an alternating polarity relative to an adjacent conductive line of the first plurality of conductive lines; a second plurality of conductive lines on the dielectric material and substantially vertically aligned with the first plurality of conductive lines, each of the second plurality of conductive lines having an alternating polarity relative to an adjacent conductive line in the second plurality of conductive lines; and a plurality of conductive pillars within the dielectric material, wherein each of the plurality of conductive pillars includes one of an upper surface interfacing with a lower surface of the second plurality of conductive lines and a lower surface interfacing with an upper surface of the first plurality of conductive lines, a vertical thickness of each conductive pillar is less than a vertical thickness between the plurality of first conductive lines and the plurality of second conductive lines, a first capacitive junction is between each conductive pillar and the first plurality of conductive lines or the second plurality of conductive lines, and a second capacitive junction is between two horizontally adjacent conductive pillars of the plurality of conductive pillars.

[0006] An aspect of the disclosure includes a method, including: forming a first conductive line within a dielectric material; forming a conductive pillar within the dielectric material over the first conductive line, wherein a vertical thickness of the conductive pillar is less than a vertical thickness of the dielectric material between the first conductive line and the second conductive line; and forming a second conductive line on the dielectric material and the conductive pillar above the

first conductive line, wherein the second conductive line is substantially aligned with the first conductive line, a first capacitive junction is between the conductive pillar and one of the first conductive line and the second conductive line, and a second capacitive junction is between the conductive pillar and a horizontally adjacent conductive pillar.

[0007] Two or more aspects described in this disclosure, including those described in this summary section, may be combined to form implementations not specifically described herein.

[0008] The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features, objects and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The embodiments of this disclosure will be described in detail, with reference to the following figures, wherein like designations denote like elements, and wherein:

[0010] FIG. 1 shows a perspective view of a structure according to embodiments of the disclosure.

[0011] FIG. 2 shows a cross-sectional view of a structure according to embodiments of the disclosure.

[0012] FIG. 3 shows a perspective view of a structure with an additional etch stop material according to embodiments of the disclosure.

[0013] FIG. 4 shows a perspective view of a structure with multiple additional etch stop materials according to embodiments of the disclosure.

[0014] FIG. 5 shows a perspective view of a structure with multiple conductive pillars according to embodiments of the disclosure.

[0015] FIG. 6 shows a perspective view of a structure with multiple conductive pillars and an additional etch stop material according to embodiments of the disclosure.

[0016] FIG. 7 shows a perspective view of a structure with multiple conductive pillars and multiple additional etch stop materials according to embodiments of the disclosure.

[0017] FIG. 8 shows a perspective view of a structure with conductive pillars on conductive lines according to embodiments of the disclosure.

[0018] FIG. 9 shows a perspective view of a structure with multiple conductive pillars on each conductive line according to embodiments of the disclosure.

[0019] FIGS. 10-12 depicts processes of a method to form capacitive junctions between a conductive line and conductive pillar according to embodiments of the disclosure.

[0020] FIGS. 13-24 depicts various processes in methods to form capacitive junctions between a conductive line and conductive pillar according to further embodiments of the disclosure.

[0021] It is noted that the drawings of the disclosure are not necessarily to scale. The drawings are intended to depict only typical aspects of the disclosure, and therefore should not be considered as limiting the scope of the disclosure. In the drawings, like numbering represents like elements between the drawings.

DETAILED DESCRIPTION

[0022] In the following description, reference is made to the accompanying drawings that form a part thereof, and in which is shown by way of illustration specific illustrative embodiments in which the present teachings may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the present teachings, and it is to be understood that other embodiments may be used and that changes may be made without departing from the scope of the present teachings. The following description is, therefore, merely illustrative.

[0023] It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or “over” another element, it may be directly on the other element or intervening

elements may also be present. In contrast, when an element is referred to as being “directly on” or “directly over” another element, there may be no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it may be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

[0024] Reference in the specification to “one embodiment” or “an embodiment” of the present disclosure, as well as other variations thereof, means that a particular feature, structure, characteristic, and so forth described in connection with the embodiment is included in at least one embodiment of the present disclosure. Thus, the phrases “in one embodiment” or “in an embodiment,” as well as any other variations appearing in various places throughout the specification are not necessarily all referring to the same embodiment. It is to be appreciated that the use of any of the following “/,” “and/or,” and “at least one of,” for example, in the cases of “A/B,” “A and/or B” and “at least one of A and B,” is intended to encompass the selection of the first listed option (a) only, or the selection of the second listed option (B) only, or the selection of both options (A and B). As a further example, in the cases of “A, B, and/or C” and “at least one of A, B, and C,” such phrasing is intended to encompass the first listed option (A) only, or the selection of the second listed option (B) only, or the selection of the third listed option (C) only, or the selection of the first and the second listed options (A and B), or the selection of the first and third listed options (A and C) only, or the selection of the second and third listed options (B and C) only, or the selection of all three options (A and B and C). This may be extended, as readily apparent by one of ordinary skill in the art, for as many items listed.

[0025] Embodiments of the disclosure provide a structure and method to provide capacitive junctions between a conductive line and conductive pillar. Structures of the disclosure include a first conductive line within a dielectric material. The dielectric material extends over the first conductive line. A second conductive line is within the dielectric material and substantially vertically aligned with the first conductive line. A conductive pillar is within the dielectric material between the first conductive line and the second conductive line. The conductive pillar includes an upper surface contacting a lower surface of the second conductive line or a lower surface contacting an upper surface of the first conductive line. A vertical thickness of the conductive pillar is less than a vertical thickness between the first conductive line and the second conductive line. A first capacitive junction is between the conductive pillar and one of the first conductive line and the second conductive line. A second capacitive junction is between the conductive pillar and a horizontally adjacent conductive pillar. As a result, portions of existing dielectric material (e.g., inter-level dielectric (ILD) layer, etch stop layer (ESL) material, etc.) provide capacitive junctions. Through embodiments of the disclosure, a circuit fabricator can omit one or more conventional capacitor structures on a device layer, as well as any intermediate isolating or intercoupling components that would otherwise be needed to provide the desired operational characteristics.

[0026] Referring to FIGS. 1 and 2, in which FIG. 1 provides a perspective view and FIG. 2 provides a cross-sectional view in plane X-Z, of a structure **100** to provide capacitive junctions between a conductive pillar and a conductive line according to embodiments of the disclosure. Structure **100** may be, e.g., within metal wiring layers (also known as “back end of line” (BEOL) layers) of a device. Structure **100** may include one or more first conductive lines **106** (four **106a-d** shown in FIG. 1, four **106a-d** shown in FIG. 2) within a dielectric material **108**. Any of the illustrative first conductive lines **106** may be referenced separately as first conductive line **106a**, **106b**, **106c**, **106d**, and are collectively referenced as “first conductive lines **106**.” Dielectric material **108** may include any currently known or later developed insulative substance, including for instance oxide-based insulators (e.g., various silicon oxide compounds), nitride based insulators, and/or other insulative materials particularly those having a dielectric constant of less than approximately 3.9.

[0027] Where structures **100** are provided within metal wiring layers of a device, dielectric material **108** may include an inter-level dielectric (ILD) layer for separating different layers from each other. In this case, portions of dielectric material **108** may include etch stop layers (ESL) **109** (shown in FIG. 2, also FIGS. 3, 4, 6, 7 discussed herein). ESL **109** may include, e.g., any material distinct from dielectric material **108** and having greater resistance to etching, to structurally distinguish between vertically adjacent wiring layers. As discussed elsewhere herein, ESLs **109** may be structurally distinct from dielectric material **108** and may be formed in a variety of configurations for additional electrical and physical separation between electrically active conductive materials. In some cases, ESL **109** may include an insulative material. In other cases, ESL **109** may include one or more conductive materials, e.g., to provide and/or maintain alternating polarity between the conductive lines **106a**, **106b**, **106c**, **106d**.

[0028] Each first conductive line **106** within dielectric material **108** may include any currently known or later developed conductive metal suitable to accumulate charge across a dielectric junction. First conductive line **106**, in one example, may include silicon-based conductors materials or other doped semiconductor conductive materials capable of operating as a capacitor terminal. In further embodiments, first conductive line **106** may include one or more metals, e.g., tungsten (W), titanium (Ti), ruthenium (Ru), cobalt (Co), copper (Cu), aluminum (Al), etc. First conductive lines **106** may be formed by depositing conductive materials on an underlying layer (e.g., another layer of dielectric material) e.g., dielectric material **108** is formed thereon. First conductive lines **106** in different positions may be electrically decoupled from each other, e.g., they may be isolated from each other by portions of dielectric material **108**. Multiple first conductive lines **106** may be present in structure **100** to provide alternating polarity terminals of distinct capacitive junctions as discussed herein.

[0029] Structure **100** also includes one or more second conductive lines **110** within dielectric material **108**. Portions of dielectric material **108** with second conductive line(s) **110** therein may constitute a different layer of dielectric material **108** from where first conductive lines **106** are positioned. Any of the illustrative second conductive lines **110** may be referenced separately as second conductive line **110a**, **110b**, **110c**, **110d**, and are collectively referenced as “second conductive lines **110**.” Second conductive lines **110** may include the same or similar example material(s) as first conductive line(s) **106**. Thus, each first conductive line **106** and each second conductive line may include Cu or any metal(s) discussed herein. Each second conductive line **110** is substantially vertically aligned with, or otherwise directly over, a respective first conductive line **106**. First conductive lines **106** and second conductive lines **110** may be separated from each other by a vertical distance that is large enough to prevent any direct coupling between conductive lines **106**, **110**, e.g., such that distinct conductive wires of a product will not electrically interfere with each other and/or produce electrical shorting as a device operates.

[0030] Embodiments of structure **100** include a set of first capacitive junctions Jc1 in selected locations by providing one or more conductive pillars **114** to first conductive line(s) **106** and/or second conductive line(s) **110** that do not completely traverse the vertical distance (of dielectric layer **108**) between conductive lines **106**, **110**. That is, each conductive pillar **114** protrudes vertically through only a portion of the vertical thickness of dielectric layer **108** between conductive lines **106**, **110** and thus does not provide a vertical interconnect (“via”) between each pair of substantially vertically aligned conductive lines **106**, **110**. Furthermore, embodiments of structure **100** also include a set of second capacitive junctions Jc2 across dielectric layer between horizontally adjacent conductive pillars **114** within the same wiring level. Conductive pillars **114** may have the same composition or similar compositions to conductive lines **106**, **110**. In the case where first conductive lines **106** and second conductive lines **110** include Cu, conductive pillars may include a metal or other conductive material composition different from Cu, e.g., other conductive semiconductor materials (e.g., doped Si or silicon germanium (SiGe), metals such aluminum, (Al), copper (Cu), zinc (Zn), gold (Au), and/or other conductive materials. Conductive

pillars **114** each may include refractory metal liners (not shown for simplicity of illustration) including, for example, ruthenium (Ru), tantalum (Ta), titanium (Ti), tungsten (W), iridium (Ir), rhodium (Rh) and platinum (Pt), etc., or mixtures of thereof. Conductive pillars **114** also may include other types of conductive materials, e.g., conductive nitride compounds (TaN, TiN, etc.). [0031] In the case of FIGS. **1** and **2**, conductive pillars **114** are below a corresponding first conductive line **106** or second conductive line **110**, but conductive pillars **114** alternatively may be in a variety of other locations as shown in FIGS. **5-9** and discussed herein. Conductive pillars **114** may be identified separately herein as **114a**, **114b**, **114c**, **114d**, **114e**, **114f**, **114g**, **114h**, or collectively as conductive pillars **114**. Each conductive pillar **114** may extend vertically through dielectric material **108** only partway between one first conductive line **106** and a vertically aligned second conductive line **110**. Second conductive lines **110** each may have a lower surface **S0** that is vertically distal to an upper surface **S1** of first conductive line **106** thereunder. Conductive pillar **114** may physically interface, i.e., electrically contact, with lower surface **S0**, and itself may have a lower surface **S2** that is between surfaces **S1**, **S2**. Thus, lower surface **S2** of each conductive pillar **114** does not contact upper surface **S1** of first conductive line **106**.

[0032] Unlike conventional via structures, conductive pillar **114** in embodiments of structure **100** does not conductively couple conductive lines **106**, **110** together. A portion of dielectric material **108** remains vertically between conductive pillar **114** and first conductive line **106**. In further embodiments discussed herein, dielectric material **108** instead may be vertically between conductive pillar **114** and second conductive line **110**, in which case conductive pillar **114** physically interfaces (electrically contacts) with upper surface **S1** of second conductive line **110**.

[0033] In structure **100**, the portion(s) of dielectric material **108** between first conductive line **106** (or second conductive line **110**) and conductive pillar **114** defines first capacitive junction **Jc1** within structure **100**. Applying a voltage differential across conductive lines **106**, **110** will induce an electric field across dielectric material **108** (and first capacitive junction **Jc1**), thereby accumulating a net charge on either first conductive line **106** or second conductive line **110** (depending on voltage polarity). During operation, first capacitive junction **Jc1** will cause charge to accumulate on any surface one of the two conductive lines **106**, **110** that is aligned with conductive pillar **114** through dielectric material **108**, thereby providing a capacitor structure for a device. Structure **100** provides multiple capacitive interfaces, in multiple directions. During operation, a second capacitive junction **Jc2** between horizontally adjacent conductive pillars conductive lines **106**, **110** and conductive pillars **114** will cause charge to accumulate on horizontally adjacent surfaces through dielectric material **108**.

[0034] Conductive lines **106**, **110** in structure **100** may have varying polarities. To function as a high density capacitor, each pair of vertically aligned conductive lines **106**, **110** has an opposite polarity. For instance, second conductive line **110a** and conductive pillar **114a** thereunder may be positively charged, in which case first conductive line **106a** therebelow is negatively charged. An adjacent second conductive line **110b** and its conductive pillar **114b** thereunder may be negatively charged and electrically separated from first conductive line **110a**. In this case, first conductive line **106b** below conductive pillar **114b** may be positively charged. A row of four second conductive lines **110a**, **110b**, **110c**, **110d** may have alternating polarities, e.g., second conductive lines **110a**, **110c** may be positively charged and second conductive lines **110b**, **110d** may be negatively charged. Conductive pillars **114a**, **114b**, **114c**, **114d** thereunder may have the same polarity as their overlying second conductive line **110a**, **110b**, **110c**, **110d** by having a conductive physical interface. First conductive lines **106a**, **106b**, **106c**, **106d** may have the opposite polarity from their vertically aligned second conductive lines **110a**, **110b**, **110c**, **110d**. In this arrangement, structure **100** may provide an alternating polarity metal-insulator-metal (APMIM) capacitor structure. Conductive lines **106**, **110** and conductive pillars **114** may provide the two conductive (“metal”) layers for each first capacitive junction **Jc1**, and dielectric material **108** may provide the insulative layer between the oppositely charged conductive (“metal”) layers. The alternating polarities of horizontally

adjacent conductive lines **110a**, **110b**, **110c**, **110d** and conductive pillars **114a**, **114b**, **114c**, **114d** also allows second capacitive junctions **Jc2** to be present therebetween as discussed herein.

[0035] To manage the current density in structure **100**, each first conductive line **106** optionally may be one of several conductive lines extending horizontally outward from a first conductive wire **120** oriented perpendicularly to first conductive lines **106** and horizontally abutting (electrically contacting) the end of multiple first conductive lines **106**. According to the illustrated example, two first conductive wires **120** (one shown in dashed lines in FIG. 1) each may abut and connect to corresponding pairs of first conductive lines **106**, i.e., lines **106a**, **106c** for one first conductive wire **120** and lines **106b**, **106d** for another first conductive wire **120**. Each conductive line **106**, **110** may extend horizontally outwardly from its respective conductive wire **106**, **110**. In this manner, each first conductive wire **120** may be coupled to similarly polarized conductive lines to preserve the alternating polarity arrangement of capacitors. In the same example, two second conductive wires **122** (one shown in dashed lines in FIG. 1) each may abut and electrically connect to corresponding pairs of second conductive lines **110**, i.e., lines **110a**, **110c** for one second conductive wire **122** and lines **110b**, **110d** for another second conductive wire **122**. To prevent electrical shorting from conductive lines **106**, **110** to conductive wires **120**, **122** having an opposite polarity (and to form second capacitive junction **Jc2** therebetween) each conductive line **106**, **110** may be horizontally displaced from the opposite polarity conductive wires **120**, **122**, e.g., along Y-axis as shown. In FIG. 2, several conductive lines **106**, **110** are shown in dashed lines to indicate that they are out of plane from the cross-sectional view shown.

[0036] Optionally, first conductive lines **106** also may include conductive pillars **114** (individually labeled **114e**, **114f**, **114g**, **114h**) thereunder. In some cases, conductive pillars **114** may not necessarily form first capacitive junction **Jc1** with any underlying structures (e.g., as shown in FIG. 1), but nonetheless may include second capacitive junction(s) **Jc2** with adjacent conductive pillars **114**. In the example of FIG. 2, however, a lowermost conductor **128** (FIG. 2 only) may be beneath conductive pillars **114** and first conductive lines **106**, thereby defining another first capacitive junction **Jc1**. Lowermost conductor **128** may not include any conductive pillars **114** coupled thereto, and simply may provide an opposite polarity conductive material beneath ESL **109**.

[0037] Turning to FIG. 3, structure **100** may include ESLs **109** within or beneath dielectric material(s) **108** in any of a variety of configurations. FIG. 3 provides an example where one ESL **109** extends beneath all of first conductive lines **106** and conductive pillars **114** thereunder, and beneath all of second conductive lines **110** and conductive pillars **114** thereunder. Each ESL **109** is shown by example as having a negligible vertical thickness, but the thickness of ESLs **109** may vary for particular applications. In some cases (e.g., as shown in FIG. 2 and discussed previously), each ESL **109** may extend vertically between lower surface **S2** of conductive pillars **114** and upper surface **S1** of first conductive lines **106**. ESL **109** also may have any desired intermediate thickness, e.g., by controlling the deposition time to form ESL **109** during manufacture. Regardless of ESL **109** thickness, ESL **109** may be in the form of a single layer extending horizontally below all conductive pillars **114**, e.g., by depositing one ESL **109** and forming conductive pillars **114** and second conductive lines **110** thereover. In this case, ESL **109** may be a dielectric material, e.g., to retain alternating polarity between the conductive pillars **114a**, **114b**, **114c**, **114d** without electrical coupling therebetween.

[0038] Turning to FIG. 4 further implementations of structure **100** may include multiple distinct ESLs **109**, each of which may be positioned beneath only one conductive pillar **114**. Each ESL **109** (separately labeled **109a**, **109b**, **109c**, **109d**, **109e**, **109f**, **109g**, **109h**) may be on dielectric material **108** and beneath a respective conductive pillar **114**. Portions of dielectric material **108** may be horizontally between each of the various ESLs **109** in structure **100**. ESLs **109** are discontinuous from each other, e.g., in implementations where ESL **109** includes a conductive material, to prevent electrical shorting between conductive pillars **114** through the composition of ESL **109**. In some implementations, the possibility of electrical shorting through ESL **109** may be negligible and thus

only one ESL **109** may be desired (e.g., as shown in FIGS. **2**, **3**). Regardless of the shape and size of ESL **109**, other aspects of structure **100** may be similar or identical to other embodiments discussed herein.

[0039] FIGS. **5-7** depict further implementations of structure **100**, in which each first conductive line **106** and each second conductive line **110** vertically interface with a plurality of pillars **144** within dielectric material **108**. Each plurality of pillars **144** may include several individual conductive pillars **114**, such that each plurality of pillars **144** is arranged as a set of horizontally distinct columns. Each conductive pillar **114** in each plurality of pillars **144** above first conductive lines **106** may physically interface (electrically connect) with lower surface **S0** of second conductive lines **110**. Each conductive pillar **114** also may have a lower surface **S2** that is above upper surface **S1** of first conductive lines **106**. Where conductive pillars **114** are provided in pluralities **144**, each conductive pillar **114** may have a horizontal length **L1** (e.g., along Y-axis as shown) that is less than a horizontal length **L2** of first conductive line **106** and/or second conductive line **110** thereon. Conductive lines **106**, **110** may be of substantially the same length or may have different lengths. Pluralities of pillars **144** may be included in structure **100**, e.g., to increase the total number of first capacitive junctions **Jc1** between first conductive lines **106** and second conductive lines **110**, as well as second capacitive junctions **Jc2** between adjacent pillars **114** with opposite polarities, which may affect the total capacitance by changing the surface area distribution of capacitive junctions **Jc1**, **Jc2**.

[0040] Pluralities of pillars **144** may be combined with other additional or alternative embodiments discussed herein. FIG. **6**, for example, includes one ESL **109** located below each plurality of pillars **144** connecting to first conductive lines **106** and another ESL **109** located below each plurality of pillars **144** connecting to second conductive lines **110**. Additionally, as shown in FIG. **7**, a larger number of separately structured ESLs **109** may be provided such that each plurality of pillars **144** is located on one respective ESL **109**. The inclusion and number of ESLs **109** may serve substantially the same technical purposes described relative to other embodiments, e.g., they may increase the capacitance between conductive pillars **114** and conductive line(s) **106**, **110**. Additionally, the use of distinct ESLs **109** for each plurality of pillars **144** may mitigate the chance of electrical shorting from one plurality of pillars **144** to an adjacent plurality of pillars **144**.

[0041] FIGS. **8** and **9** depict still further embodiments of structure **100** in which conductive pillars **114** are positioned on conductive lines **106**, **110**, such that first capacitive junction **Jc1** is vertically between lower surface **S0** of second conductive line **110** and an upper surface **S4** of one or more conductive pillars **114** thereunder. In this case, a lower surface **S3** of each conductive pillar **114** may be on (i.e., land on and electrically connect with) a respective first conductive line **106** or second conductive line **110**. The vertical separation distance between lower surface **S0** of second conductive line **110** and upper surface **S4** of conductive pillar(s) **114** may be less than the distance between lower surface **S3** of any conductive pillar(s) **114** on second conductive line **110** and upper surface **S4** of underlying conductive pillar(s) **114**. No direct physical interface is present between second conductive line **110** and any conductive pillar(s) **114** on first conductive line **106**. Thus, first capacitive junction **Jc1** is present between conductive lines **110** and pillars **114** instead of a direct physical and electrical interface between the two lines **106**, **110** through conductive pillar(s). In addition, second capacitive junction **Jc2** remains present between horizontally adjacent oppositely charged pillars **114** and conductive lines **106**, **110** in the same layer through dielectric material **108**.

[0042] Despite the change in position for conductive pillar(s) **114** relative to conductive line(s) **106**, **110**, structure **100** may be similar to or the same as other embodiments provided herein. For instance, conductive wires **120**, **122** may extend perpendicularly relative to conductive line(s) **106**, **110** and may be coupled thereto to provide a stem and leaf arrangement or “finger” arrangement of capacitors. The use of multiple conductive wires **120**, **122** displaced from each other may allow each adjacent conductive line **106**, **110** to have an alternating polarity as discussed elsewhere herein. Additionally, as shown in FIG. **9**, conductive pillar(s) **114** may be arranged in pluralities

over each conductive line **106**, **110** similar to other arrangements discussed herein and shown in FIGS. 5-7. ESL(s) also may be present within each first capacitive junction **Jc1**, and may be directly beneath each conductive line **106**, **110**. Although one ESL **109** is shown as an example in FIGS. 8 and 9, multiple ESLs **109** may be included such that each conductive line **106**, **110** is located on a respective ESL **109** as provided in other embodiments and discussed herein. ESL **109**, in further implementations, may be omitted from any embodiment of structure **100** disclosed herein (e.g., those shown in FIGS. 6-9).

[0043] Turning to FIG. 10, the disclosure also include methods to form embodiments of structure **100**. FIGS. 10-12 may be implemented to form structure **100** as shown in FIGS. 8, e.g., with conductive pillar(s) **114** on upper surfaces of first conductive lines **106**. However, the same processes may be implemented and the resulting structure may be turned upside-down to yield embodiments of structure **100** discussed elsewhere herein, e.g., with second conductive lines **110** directly on, or otherwise physically interfacing with, conductive pillar(s) **114** at their upper surfaces. An initial structure **158** to be processed may include first conductive line **106** (or second conductive line **110**) as layers of conductive material (e.g., Cu as discussed herein) within dielectric material **108** and without any pillars coupled thereto. Structure **158** may include an etch stop material **160** (e.g., a nitride-based insulator and/or other insulative materials) on dielectric material **108** and formed over first conductive lines **106** by deposition to any desired thickness. Etch stop material **160** may have an initial thickness that is much larger than the eventual thickness of ESL(s) **109** (FIGS. 2-4, 6-9) in structure **100**. Methods of the disclosure may include forming a mask **162** on etch stop material **160**, with the size and shape of mask **162** being selected to control the length and width of conductive pillar(s) **114** (FIGS. 1-9) to be formed. Mask **162** may include a developable organic planarization layer (OPL) on the layer to be etched, a developable anti-reflective coating (ARC) layer on the developable OPL, and a photoresist mask layer on the developable ARC layer. In certain implementations, it may be beneficial for the ESL **109** to have a vertical thickness significantly smaller than the vertical thickness of dielectric material **108**, e.g., to prevent planarity disturbance and to reduce processing complexity.

[0044] Turning to FIGS. 11-13, further processing may include removing (e.g., by etching) any portions of etch stop material **160** (FIG. 10) not covered by mask **162** (FIG. 10). Mask **162** then can be removed (e.g., by stripping). Portions of etch stop material **160** then can be removed (e.g., by etching), e.g., as shown in FIG. 12. As shown in FIG. 12, a remaining portion of etch stop material **160** remains as ESL **109** on conductive line **106**, **110**. FIG. 12 also depicts forming an additional portion of dielectric material **108** over ESL **109**.

[0045] Referring to FIG. 13, further processing may include forming openings **Y** within dielectric material **108**, in which ESL **109** prevents openings **Y** from extending completely through dielectric materials **108** to expose first conductive lines **106**.

[0046] FIG. 14 depicts forming conductive pillar(s) **114**, and second conductive lines **110**, within openings **Y**. The forming of conductive pillars **114** and second conductive lines **110** may be by deposition and etching of conductive material within openings **Y**, and filling (with etch back and/or planarization) of openings **Y** to form second conductive lines **110** and conductive pillars **114**, i.e., they may be formed together. Conductive pillars **114** thus may be vertically above first conductive line **106** and first capacitive junction **Jc1** is between each first conductive line **106** and each conductive pillar **114**. Second capacitive junctions **Jc2** may be horizontally between adjacent conductive pillars **114**. Subsequent processing may include, e.g., forming additional conductive line(s) **106**, **110**, conductive pillar(s) **114** and ESLs **109** over second conductive line **110** to provide additional layers with additional capacitive junctions.

[0047] FIGS. 15-17 depict further methods according to the disclosure in which preliminary structure **158** initially includes ESL **109** on conductive line(s) **106**, **110** and dielectric material **108** on ESL **109**, instead of a single thicker layer of ESL **109**. For certain implementations, it may be beneficial to embed the ESL **109** within dielectric material **108** (e.g., within an ILD layer). Such an

implementation is shown in FIGS. 15-17 through a portion of dielectric material **108** under ESL **109** and indicated with dashed lines. Embedding of ESL **109** within dielectric material **108** may reduce capacitance variability and, in some cases, may help to resist dielectric breakdown during operation. By this approach, methods of the disclosure include forming an opening **164** (FIG. 16 only) within dielectric material **108** (e.g., by etching with mask **162** in place) to the depth of ESL **109**. Opening **164** then may be filled with conductive material to provide conductive pillar **114**. Any excess material outside opening **164** can then be removed (e.g., by chemical mechanical planarization (CMP) or similar techniques, and another conductive line **106**, **110** then may be formed on conductive pillar **114**. The example implementation shown in FIGS. 15-17 may be suitable for embodiments of structure **100** where second conductive line **110** is on an upper surface of conductive pillar **114**. Alternatively, products created by the implementation shown in FIGS. 15-17 may be turned upside down to provide an embodiment of structure where conductive pillar **114** is on an upper surface of first conductive line **106**.

[0048] Referring now to FIGS. 18-21, methods of the disclosure may include changing the number of masks **162** on etch stop material **160**, e.g., to provide a plurality of conductive pillars **144** (FIGS. 14, 15) on conductive line(s) **106**. FIGS. 18-21 illustrate forming structure **100** as shown in FIGS. 9, e.g., with plurality of conductive pillar(s) **144** on upper surfaces of first conductive lines **106**. However, the same processes may be implemented and the resulting structure may be turned upside-down to yield embodiments of structure **100** discussed elsewhere herein, e.g., with second conductive lines **110** directly on, or otherwise physically interfacing with, plurality of conductive pillar(s) **144** at upper surface of each conductive line **106**. A similar or identical set of processing steps may be implemented with additional masks **162** with different horizontal dimensions to form any desired number of conductive pillars **114** on conductive line(s) **106**, **110**. Collectively, the conductive pillars **114** provide plurality of pillars **144**. As discussed elsewhere herein, individual ESLs **109** may be formed e.g., by removing only a portion of etch stop material **160** on dielectric material **108**. The remaining etch stop material **160** thereby provides ESL **109** on dielectric material **108**. Conductive pillars **114** then may be formed, e.g., by forming dielectric material **108** over ESLs **109**, forming openings (not shown) therein to expose ESLs **109**, and forming conductive materials within the openings to form conductive pillars **114** and second conductive lines **110**. These processes may be implemented as many times as desired to provide first capacitive junctions **Jc1** between successive metal wiring layers. In still further embodiments, the size and number of masks(s) **162** may be changed in each implementation to provide a different shape, size, or number of conductive pillar(s) **114** and/or ESLs **109** in successive layers.

[0049] FIGS. 22-24 depict further methods according to the disclosure in which preliminary structure **158** initially includes ESL **109** on conductive line(s) **106**, **110** and dielectric material **108** on ESL **109**, instead of a single thicker layer of ESL **109**. By this approach, methods of the disclosure include forming multiple openings **164a**, **164b**, **164c**, **164d** (FIG. 23 only) within dielectric material **108** (e.g., by etching with mask **162** in place) to the depth of ESL **109**. Opening **164** then may be filled with conductive material to provide plurality of conductive pillars **144**. Any excess material outside openings **164a**, **164b**, **164c**, **164d** can then be removed (e.g., by chemical mechanical planarization (CMP) or similar techniques, and another conductive line **106**, **110** then may be formed on plurality of conductive pillars **144**. The example implementation shown in FIGS. 22-24 implementation may be for embodiments of structure **100** where second conductive line **110** is on an upper surface of plurality of conductive pillars **144**. Alternatively, in such an implementation, the resulting product may be turned upside down to provide an embodiment of structure where plurality of conductive pillars **144** is on an upper surface of first conductive line **106**.

[0050] Embodiments of the disclosure provide various technical and commercial advantages, examples of which are discussed herein. Among other things, structure **100** may provide improvements in capacitive density as compared to conventional capacitors formed in metal wiring

layers and/or device layers. In some cases, e.g., where pluralities of pillars **144** are present between conductive lines **106, 110**, the capacitive density may improve by sixty percent or more relative to conventional devices. In turn, this reduces surface area required to provide a desired number of capacitors within a device structure. In addition to these and other technical benefits, embodiments of the disclosure are capable of providing APMIM capacitor arrangements simply by including multiple conductive wires **120, 122** coupled to alternating conductive lines **106, 110** in structure **100**.

[0051] The structure and method as described herein are used in the fabrication of integrated circuit chips. The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case the chip is mounted in a single chip package (such as a plastic carrier, with leads that are affixed to a motherboard or other higher-level carrier) or in a multichip package (such as a ceramic carrier that has either or both surface interconnections or buried interconnections). In any case the chip is then integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either (a) an intermediate product, such as a motherboard, or (b) an end product. The end product can be any product that includes integrated circuit chips, ranging from toys and other low-end applications to advanced computer products having a display, a keyboard or other input device, and a central processor.

[0052] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the,” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. “Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur, and that the description includes instances where the event occurs and instances where it does not.

[0053] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. “Approximately” as applied to a particular value of a range applies to both values, and unless otherwise dependent on the precision of the instrument measuring the value, may indicate $\pm 10\%$ of the stated value(s).

[0054] The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present disclosure has been presented for purposes of illustration and description but is not intended to be exhaustive or limited to the disclosure in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The embodiment was chosen and described in order to best explain the principles of the disclosure and the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

Claims

1. A structure comprising: a first conductive line within a dielectric material, the dielectric material extending over the first conductive line; a second conductive line within the dielectric material and substantially vertically aligned with the first conductive line; and a conductive pillar within the dielectric material between the first conductive line and the second conductive line, wherein the conductive pillar includes one of an upper surface contacting a lower surface of the second conductive line and a lower surface contacting an upper surface of the first conductive line, wherein a vertical thickness of the conductive pillar is less than a vertical thickness between the first conductive line and the second conductive line, a first capacitive junction is between the conductive pillar and one of the first conductive line and the second conductive line.
2. The structure of claim 1, wherein the first conductive line and the second conductive line are each one of a plurality of conductive lines extending horizontally outward from a conductive wire.
3. The structure of claim 1, wherein the conductive pillar is one of a plurality of pillars within the dielectric material, each of the plurality of pillars having a horizontal length less than a horizontal length of the first conductive line or the second conductive line.
4. The structure of claim 1, wherein the dielectric material includes an etch stop layer (ESL) vertically between the conductive pillar and one of the first conductive line and the second conductive line.
5. The structure of claim 4, wherein the ESL vertically interfaces with the conductive pillar and each of a plurality of additional conductive pillars between the first conductive line and the second conductive line.
6. The structure of claim 4, wherein the dielectric material further includes an inter-level dielectric (ILD) layer on the ESL, the ILD layer having a different composition from the ESL.
7. The structure of claim 6, wherein the ILD layer includes silicon carbonitride (SiCN).
8. The structure of claim 1, wherein a second capacitive junction is between the conductive pillar and a horizontally adjacent conductive pillar.
9. A structure comprising: a first plurality of conductive lines within a dielectric material, each of the first plurality of conductive lines having an alternating polarity relative to an adjacent conductive line of the first plurality of conductive lines; a second plurality of conductive lines on the dielectric material and substantially vertically aligned with the first plurality of conductive lines, each of the second plurality of conductive lines having an alternating polarity relative to an adjacent conductive line in the second plurality of conductive lines; and a plurality of conductive pillars within the dielectric material, wherein each of the plurality of conductive pillars includes one of an upper surface interfacing with a lower surface of the second plurality of conductive lines and a lower surface interfacing with an upper surface of the first plurality of conductive lines, a vertical thickness of each conductive pillar is less than a vertical thickness between the plurality of first conductive lines and the plurality of second conductive lines, a first capacitive junction is between each conductive pillar and one of the first plurality of conductive lines and the second plurality of conductive lines, and a second capacitive junction is between two horizontally adjacent conductive pillars of the plurality of conductive pillars.
10. The structure of claim 9, further comprising: a first conductive wire coupled to the first plurality of conductive lines, wherein each of the first plurality of conductive lines extends horizontally outwardly from the first conductive wire; and a second conductive wire coupled to the second plurality of conductive lines, wherein each of the second plurality of conductive lines extends horizontally outwardly from the second conductive wire.
11. The structure of claim 9, wherein the dielectric material includes an etch stop layer (ESL) vertically between each of the plurality of conductive pillars and the respective first conductive line or the respective second conductive line.

- 12.** The structure of claim 11, wherein the dielectric material further includes an inter-level dielectric (ILD) layer on the ESL, the ILD layer having a different composition from the ESL.
- 13.** The structure of claim 12, wherein the ILD layer includes silicon carbonitride (SiCN).
- 14.** The structure of claim 9, wherein a material composition of the plurality of conductive pillars is different from a material composition of the first plurality of conductive lines or the second plurality of conductive lines.
- 15.** A method comprising: forming a first conductive line within a dielectric material; forming a conductive pillar within the dielectric material over the first conductive line, wherein a vertical thickness of the conductive pillar is less than a vertical thickness of the dielectric material between the first conductive line and the second conductive line; and forming a second conductive line on the dielectric material and the conductive pillar above the first conductive line, wherein the second conductive line is substantially aligned with the first conductive line, a first capacitive junction is between the conductive pillar and one of the first conductive line and the second conductive line.
- 16.** The method of claim 15, wherein a second capacitive junction is between the conductive pillar and a horizontally adjacent conductive pillar.
- 17.** The method of claim 15, wherein forming the conductive pillar includes forming a plurality of pillars within the dielectric material, each of the plurality of pillars having a horizontal length less than a horizontal length of the first conductive line or the second conductive line.
- 18.** The method of claim 15, wherein forming the dielectric material includes forming an etch stop layer (ESL) vertically between the conductive pillar and one of the first conductive line and the second conductive line.
- 19.** The method of claim 18, wherein the ESL vertically interfaces with the conductive pillar and each of a plurality of additional conductive pillars between the first conductive line and the second conductive line.
- 20.** The method of claim 18, wherein forming the dielectric material further includes forming an inter-level dielectric (ILD) layer on the ESL, the ILD layer having a different composition from the ESL.
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